THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A man stands 3 m in front of a plane mirror and a boy stands 2 m behind the man. The distance of the boy's image from the man is: 1

(a) 5 m **(b)** 3 m **(c)** 8 m **(d)** 2 m

2. A total lunar eclipse is seen at midnight from a place. At that time the Moon is: 1

(a) close to the Sun **(b)** below the horizon
(c) high in the sky **(d)** not present in the sky

3. If a pesticide kills all the frogs in a field, the number of insects will most likely: 1

(a) increase **(b)** decrease **(c)** become constant **(d)** stay zero

4. A student says the Earth's inner core is liquid because it is hotter than the outer core. This reasoning is wrong because: 1

(a) pressure raises the melting point **(b)** iron cannot be solid
(c) the inner core is gas **(d)** the outer core is cooler

5. A student heats three metal rods of equal size and finds wax dropping at different times. The dependent variable is: 1

(a) time for wax to drop **(b)** flame
(c) metal type **(d)** length of rod

6. Which is NOT a feature of a unicellular organism? 1

(a) nucleus (in protozoa) **(b)** cytoplasm
(c) tissues **(d)** cell membrane

7. A new disease spreads from person to person. Which first step reduces the spread? 1

(a) closing hospitals **(b)** isolating the patients and hygiene
(c) stopping all food **(d)** ignoring symptoms

8. A student's 'magnet' was a steel rod in a coil and kept its magnetism after the current stopped. This shows that steel: 1

(a) retains magnetism **(b)** loses magnetism quickly
(c) is an insulator **(d)** cannot be magnetised

9. To reduce the wear of the surfaces of the moving parts of a machine, we should: 1

(a) increase the load **(b)** apply a lubricant
(c) use a rough material **(d)** stop the machine

10. A coastal village plants mangroves to: 1

(a) create wind **(b)** stop wind completely
(c) reduce the force of the surge **(d)** increase rain

11. Which has the weakest forces between its particles at room temperature? 1

(a) ice **(b)** oxygen **(c)** iron **(d)** water

12. A shiny, hard element conducts electricity and turns black in air with sulphur compounds. It could be: 1

(a) sulphur **(b)** carbon **(c)** neon **(d)** silver

13. A sample of tap water leaves white deposits on evaporation. This shows that it contains: 1

(a) oil **(b)** dissolved solids
(c) pure water **(d)** dissolved gases only

14. A convex mirror is fitted in shops to see a wide area because it forms: 1

(a) an inverted image of the same size
(b) no image
(c) an erect, diminished image that covers a wide field
(d) an enlarged real image

15. After 33 years the lunar and solar years differ by about: 1

(a) one year **(b)** one month **(c)** ten years **(d)** one day

16. Lichen is an association of: 1

(a) algae and animals **(b)** bacteria and plants
(c) algae and fungi **(d)** fungi and animals

17. A town draws more water from wells than rain recharges. In the long term: 1

(a) ozone increases **(b)** the water table rises
(c) wells overflow **(d)** the water table falls

18. Assertion (A): Records should include observations that do not match the expectation. Reason (R): Selective reporting gives an incomplete and misleading picture. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. Assertion (A): A red blood cell can carry more oxygen because it has no nucleus. Reason (R): Haemoglobin takes up the space gained. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. Assertion (A): Diabetes is a communicable disease. Reason (R): Diabetes is not caused by any pathogen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

21. Assertion (A): A compass needle placed under a current-carrying wire deflects. Reason (R): Electric current produces a magnetic field around the wire. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

22. Assertion (A): The weight of an object is less on the Moon than on the Earth. Reason (R): The Moon has weaker gravity than the Earth. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

23. Assertion (A): The pressure exerted by a person is more when standing on one foot. Reason (R): The area of contact is half. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

24. Assertion (A): A steam burn is more severe than a burn from boiling water. Reason (R): Steam gives out extra heat on condensing. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

25. Assertion (A): Graphite conducts electricity although carbon is a non-metal. Reason (R): Graphite is an exception among non-metals. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

26. Assertion (A): The mass of a solution is equal to the sum of the masses of solute and solvent. Reason (R): Mass is conserved during dissolving. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

27. Assertion (A): A convex lens can start a fire using sunlight. Reason (R): It converges parallel rays to a point. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

28. Assertion (A): The Moon is visible at night. Reason (R): The Moon produces its own light. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

29. Assertion (A): Decomposers are essential for the recycling of nutrients. Reason (R): They convert dead organic matter into simple inorganic substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

30. Assertion (A): The Moon has large changes in surface temperature between day and night. Reason (R): The Moon has a thick atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

31. A pendulum completes 30 swings in 48 s. The time period is: 1

(a) 78 s **(b)** 0.625 s **(c)** 16 s **(d)** 1.6 s

32. Which sequence correctly lists in increasing size: red blood cell (7 micrometres), bacterium (2 micrometres), virus (0.1 micrometre)? 1

(a) red blood cell, bacterium, virus **(b)** virus, red blood cell, bacterium
(c) virus, bacterium, red blood cell **(d)** bacterium, virus, red blood cell

33. A sick person drinks water from a well that is next to a leaking toilet pit. The disease most likely to spread is: 1

(a) typhoid **(b)** rickets **(c)** goitre **(d)** scurvy

34. A device connected to a 5 A fuse draws 3 A and a second device draws 4 A. If both are on in the same circuit, the fuse will: 1

(a) melt **(b)** become magnetic **(c)** remain intact **(d)** cool

35. When two forces F1 = 5 N (right) and F2 = 8 N (left) act on a body, a third force of what size and direction makes the net force zero? 1

(a) 13 N to the left **(b)** 3 N to the left
(c) 3 N to the right **(d)** 13 N to the right

36. A man of weight 800 N exerts 40000 Pa when wearing shoes. The area of contact of both shoes is: 1

(a) 0.2 square metre **(b)** 2 square metres
(c) 32 square metres **(d)** 0.02 square metre

37. Which of these is the best explanation for the cooling in a desert cooler? 1

(a) pads produce cold
(b) fan makes cold
(c) water is cold
(d) water evaporates and takes heat from the air

38. Which of the following represents 3 molecules of oxygen? 1

(a) $3O_2$ **(b)** $6O$ **(c)** O_6 **(d)** $3\,O$

39. How much water must be added to 100 g of a 40 per cent solution to dilute it to 10 per cent? 1

(a) 300 g **(b)** 100 g **(c)** 30 g **(d)** 400 g

40. A clock's image in a plane mirror shows 10:10 in a mirror. The actual time is: 1

(a) 11:50 **(b)** 10:10 **(c)** 2:50 **(d)** 1:50

41. The new moon of 1 January 2024 is followed by a full moon after about: 1

(a) 15 days **(b)** 30 days **(c)** 7 days **(d)** 60 days

42. A lake in which the algae grow excessively after fertiliser run-off will show: 1

(a) cool water
(b) a decrease in dissolved oxygen after the algae die
(c) clear water
(d) increase in fishes

43. A planet has a thick CO2 atmosphere and a surface hot enough to melt lead. This planet is most like: 1

(a) Venus **(b)** the Moon **(c)** Mars **(d)** Earth

44. Which is the best question to start a study of how the volume of water affects the time to boil? 1

(a) Does 200 mL of water boil faster than 400 mL on the same burner?
(b) Where do we boil water?
(c) Is boiled water good?
(d) Who boils water?

45. Why can't antibiotic medicines treat the flu? 1

(a) Antibiotics are weak **(b)** Flu is caused by a protozoan
(c) Flu is not a disease **(d)** Flu is caused by a virus

46. Why can dengue be spread even in clean-looking colonies? 1

(a) It is non-communicable
(b) Dengue is airborne
(c) Dirty water causes it
(d) Aedes breeds in small collections of clean water

47. The same heater is run for 2 hours instead of 1 hour. The heat produced is: 1

(a) zero **(b)** halved **(c)** doubled **(d)** same

48. A pair of identical rods, A and B, are charged by rubbing against different materials. They repel each other. This means: 1

(a) both have the same sign of charge **(b)** one is neutral
(c) both have opposite charges **(d)** both are magnets

49. Why does air pressure fall as altitude increases? 1

(a) less air lies above **(b)** temperature rises
(c) air is heavier **(d)** water vapour falls

50. The best way to dry a wet handkerchief quickly is to: 1

(a) keep in a damp room **(b)** dip it in cold water
(c) fold it and keep in a closed box **(d)** spread it open in warm, dry, windy place

END OF QUESTION PAPER **Total: 50 Marks**

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